

ENV 790.30 - Time Series Analysis for Energy Data | Spring 2025

Assignment 2 - Due date 01/27/26

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Submission Instructions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., “LuanaLima_TSA_A02_Sp26.Rmd”). Then change “Student Name” on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

R packages

R packages needed for this assignment: “forecast”, “tseries”, and “dplyr”. Install these packages, if you haven’t done yet. Do not forget to load them before running your script, since they are NOT default packages.\

```
## Registered S3 method overwritten by 'quantmod':
##   method                from
##   as.zoo.data.frame zoo

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

Data set information

Consider the data provided in the spreadsheet “Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source” on our **Data** folder. The data comes from the US Energy Information and Administration and corresponds to the December 2025 Monthly Energy Review. The spreadsheet is ready to be used. Refer to the file “M2_ImportingData_XLSX.Rmd” in our Lessons folder for instructions on how to read .xlsx files.

```
#Importing data set without change the original file using read.xlsx
energy_data1 <- read_excel(path="./Data/Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source")
```

```
## New names:
## * ' ' -> '...1'
## * ' ' -> '...2'
## * ' ' -> '...3'
## * ' ' -> '...4'
## * ' ' -> '...5'
## * ' ' -> '...6'
## * ' ' -> '...7'
## * ' ' -> '...8'
## * ' ' -> '...9'
## * ' ' -> '...10'
## * ' ' -> '...11'
## * ' ' -> '...12'
## * ' ' -> '...13'
## * ' ' -> '...14'
```

```
#Now let's extract the column names from row 11
read_col_names <- read_excel(path="./Data/Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source", sheet=10)
```

```
## New names:
## * ' ' -> '...1'
## * ' ' -> '...2'
## * ' ' -> '...3'
## * ' ' -> '...4'
## * ' ' -> '...5'
## * ' ' -> '...6'
## * ' ' -> '...7'
## * ' ' -> '...8'
## * ' ' -> '...9'
## * ' ' -> '...10'
## * ' ' -> '...11'
## * ' ' -> '...12'
## * ' ' -> '...13'
## * ' ' -> '...14'
```

```
#Assign the column names to the data set
colnames(energy_data1) <- read_col_names
```

```
#Visualize the first rows of the data set
head(energy_data1)
```

```
## # A tibble: 6 x 14
##   Month      'Wood Energy Production' 'Biofuels Production'
##   <dtm>                <dbl> <chr>
## 1 1973-01-01 00:00:00          130. Not Available
## 2 1973-02-01 00:00:00          117. Not Available
## 3 1973-03-01 00:00:00          130. Not Available
## 4 1973-04-01 00:00:00          125. Not Available
## 5 1973-05-01 00:00:00          130. Not Available
```

```
## 6 1973-06-01 00:00:00          125. Not Available
## # i 11 more variables: 'Total Biomass Energy Production' <dbl>,
## #   'Total Renewable Energy Production' <dbl>,
## #   'Hydroelectric Power Consumption' <dbl>,
## #   'Geothermal Energy Consumption' <dbl>, 'Solar Energy Consumption' <chr>,
## #   'Wind Energy Consumption' <chr>, 'Wood Energy Consumption' <dbl>,
## #   'Waste Energy Consumption' <dbl>, 'Biofuels Consumption' <chr>,
## #   'Total Biomass Energy Consumption' <dbl>, ...
```

Question 1

You will work only with the following columns: Total Biomass Energy Production, Total Renewable Energy Production, Hydroelectric Power Consumption. Create a data frame structure with these three time series only. Use the command `head()` to verify your data.

```
#variables I want
selected_vars <- energy_data1[,c("Month", "Total Biomass Energy Production", "Total Renewable Energy Production", "Hydroelectric Power Consumption")]

#turn into a dataframe
energy_df <- as.data.frame(selected_vars)
head(energy_df)
```

```
##      Month Total Biomass Energy Production Total Renewable Energy Production
## 1 1973-01-01          129.787          219.839
## 2 1973-02-01          117.338          197.330
## 3 1973-03-01          129.938          218.686
## 4 1973-04-01          125.636          209.330
## 5 1973-05-01          129.834          215.982
## 6 1973-06-01          125.611          208.249
##      Hydroelectric Power Consumption
## 1              89.562
## 2              79.544
## 3              88.284
## 4              83.152
## 5              85.643
## 6              82.060
```

Question 2

Transform your data frame in a time series object and specify the starting point and frequency of the time series using the function `ts()`.

```
ts_energy <- ts(energy_df[,2:4], start=c(1973,1),frequency = 12)
head(ts_energy)
```

```
##      Total Biomass Energy Production Total Renewable Energy Production
## Jan 1973          129.787          219.839
## Feb 1973          117.338          197.330
## Mar 1973          129.938          218.686
## Apr 1973          125.636          209.330
## May 1973          129.834          215.982
```

```
## Jun 1973      125.611      208.249
##      Hydroelectric Power Consumption
## Jan 1973      89.562
## Feb 1973      79.544
## Mar 1973      88.284
## Apr 1973      83.152
## May 1973      85.643
## Jun 1973      82.060
```

Question 3

Compute mean and standard deviation for these three series.

```
#means
mean(ts_energy[,1])
```

```
## [1] 286.0489
```

```
mean(ts_energy[,2])
```

```
## [1] 409.1952
```

```
mean(ts_energy[,3])
```

```
## [1] 79.35682
```

```
#standard deviations
sd(ts_energy[,1])
```

```
## [1] 96.21209
```

```
sd(ts_energy[,2])
```

```
## [1] 151.4223
```

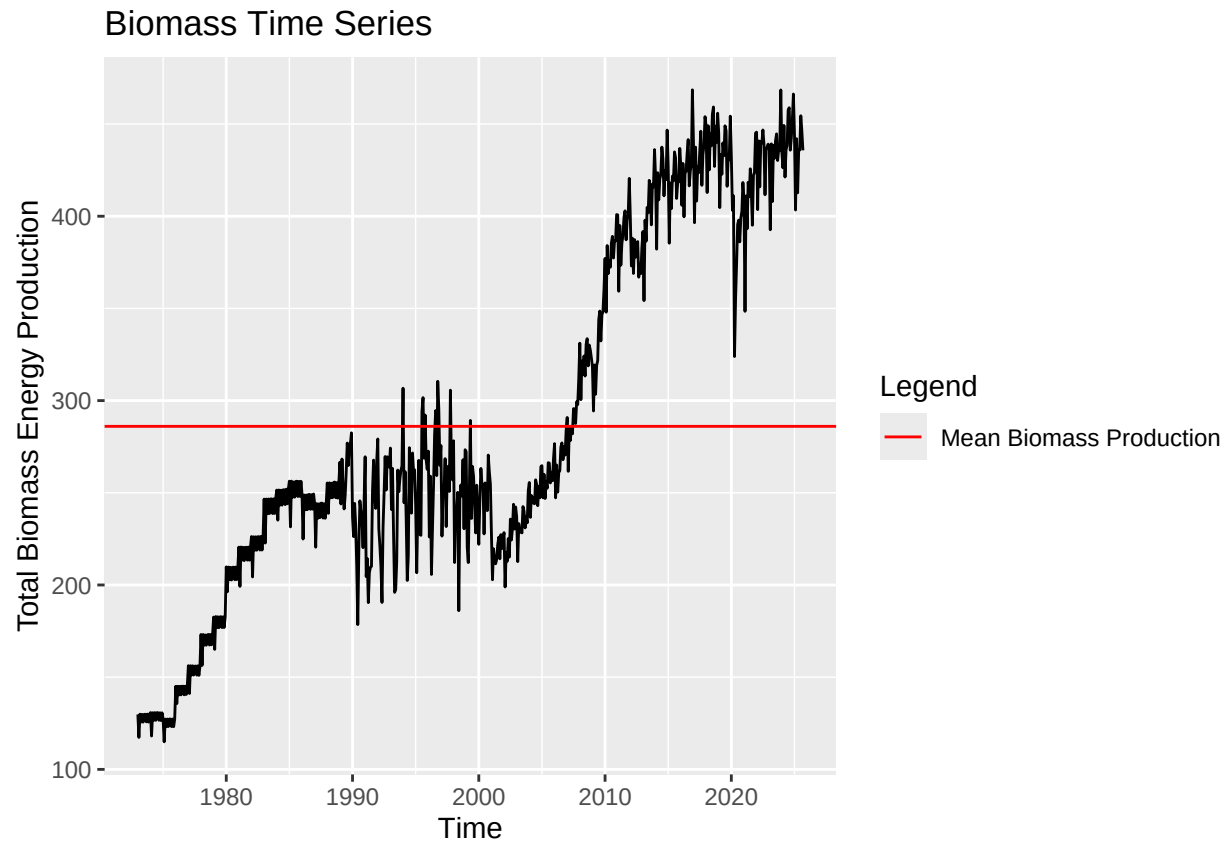
```
sd(ts_energy[,3])
```

```
## [1] 14.1202
```

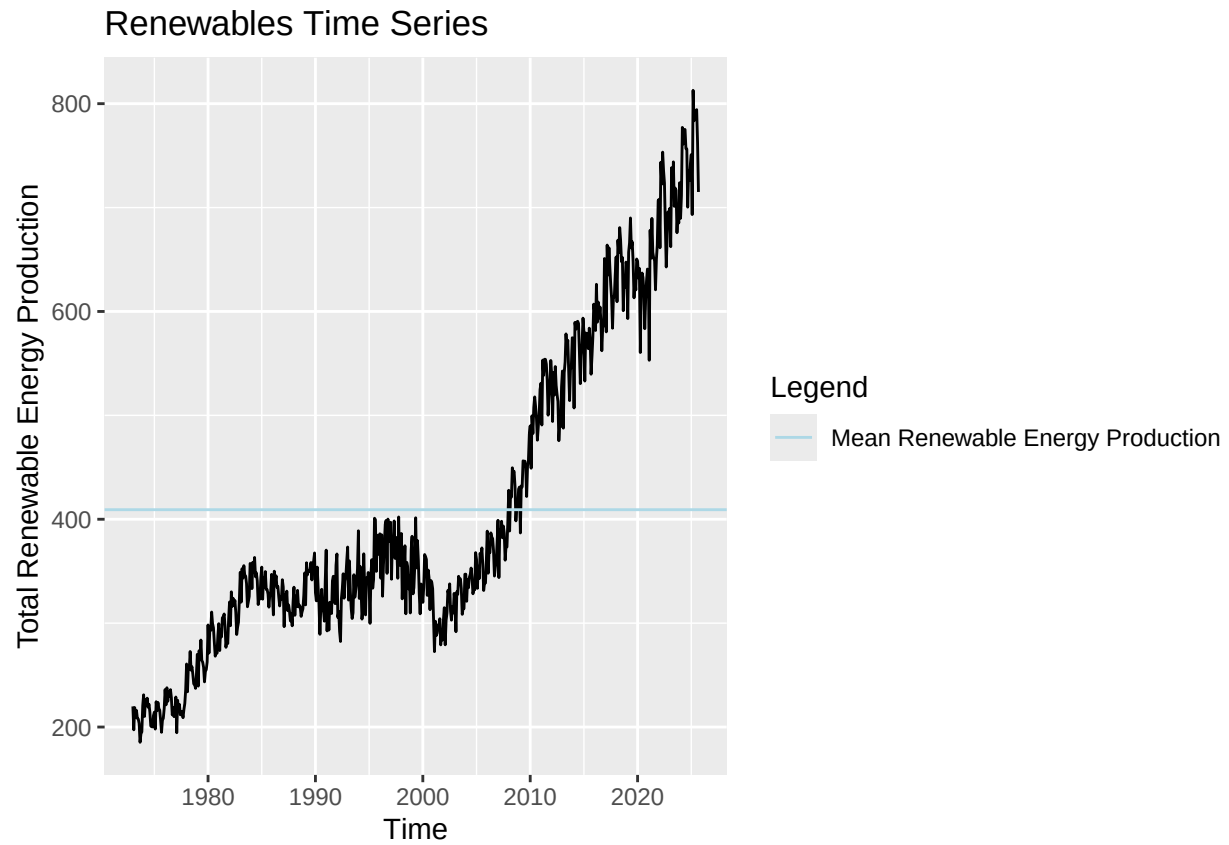
Question 4

Display and interpret the time series plot for each of these variables. Try to make your plot as informative as possible by writing titles, labels, etc. For each plot add a horizontal line at the mean of each series in a different color.

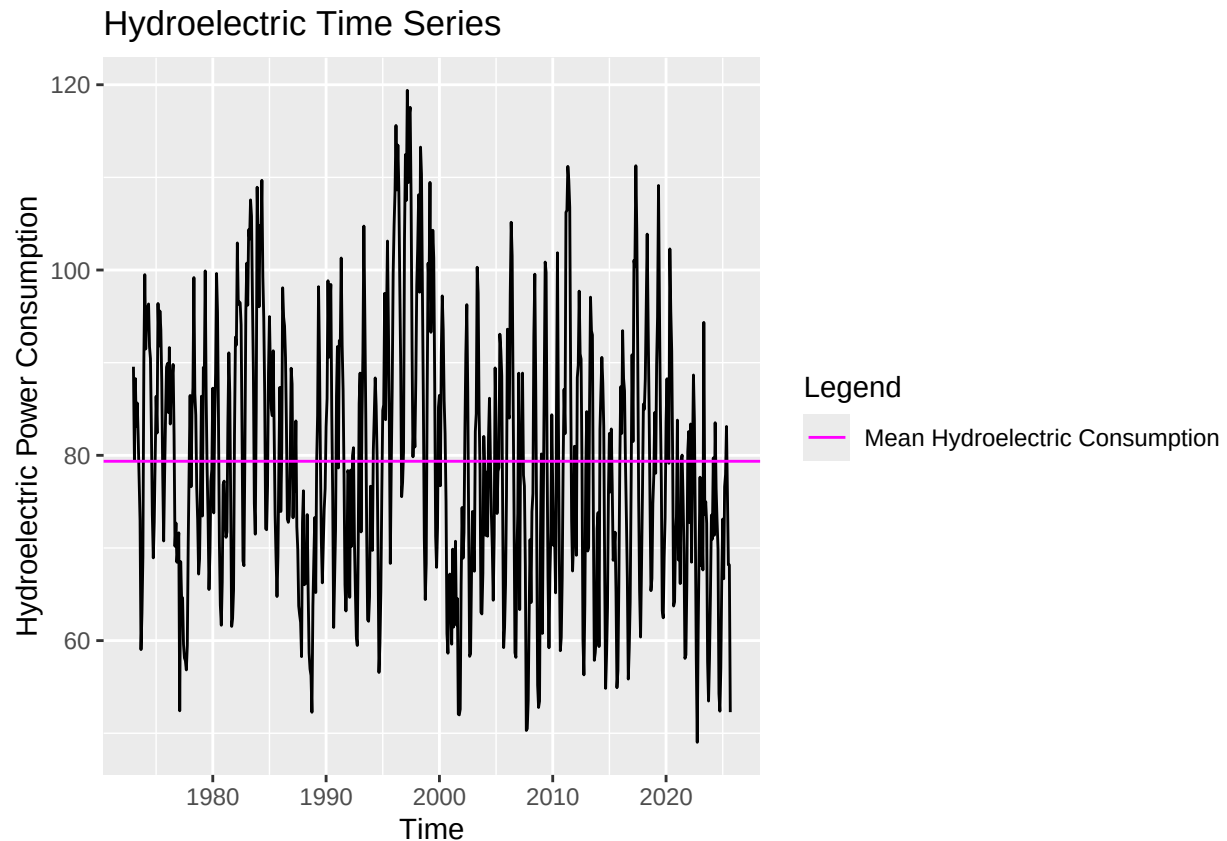
```
#first plot
autoplot(ts_energy[,1]) +
  geom_hline(aes(yintercept =mean(ts_energy[,1]), linetype ="Mean Biomass Production"), color = "red")+
  ggtitle("Biomass Time Series")+
  xlab("Time") +
  ylab("Total Biomass Energy Production")+
  labs(linetype = "Legend")
```



```
#second plot
autoplot(ts_energy[,2]) +
  geom_hline(aes(yintercept =mean(ts_energy[,2]), linetype= "Mean Renewable Energy Production"), color = "red")+
  ggtitle("Renewables Time Series")+
  xlab("Time") +
  ylab("Total Renewable Energy Production") +
  labs(linetype = "Legend")
```



```
#third plot
autoplot(ts_energy[,3]) +
  geom_hline(aes(yintercept =mean(ts_energy[,3]), linetype ="Mean Hydroelectric Consumption"), color="m") +
  ggtitle("Hydroelectric Time Series")+
  xlab("Time") +
  ylab("Hydroelectric Power Consumption") +
  labs(linetype = "Legend")
```



#not sure how to add the mean lines in an easier way.. had to do a lot of internet searching to figure

The three time series look very different from each other over time. Biomass increases, somewhat linearly, but not linearly for a decade between 1990 - 2000. Renewable production has always increased but pretty much linearly so from 2000 onwards. The hydroelectric consumption is less easy to interpret. Seems to oscillate (maybe yearly high and low). seems to fluctuate over the entire time around the mean (unlike the other two series).

Question 5

Compute the correlation between these three series. Are they significantly correlated? Explain your answer.

```
#computing correlation with function cor()
cor(ts_energy)
```

```
##                                Total Biomass Energy Production
## Total Biomass Energy Production      1.0000000
## Total Renewable Energy Production    0.9652985
## Hydroelectric Power Consumption      -0.1347374
##                                Total Renewable Energy Production
## Total Biomass Energy Production      0.96529851
## Total Renewable Energy Production    1.00000000
## Hydroelectric Power Consumption      -0.05842436
##                                Hydroelectric Power Consumption
```

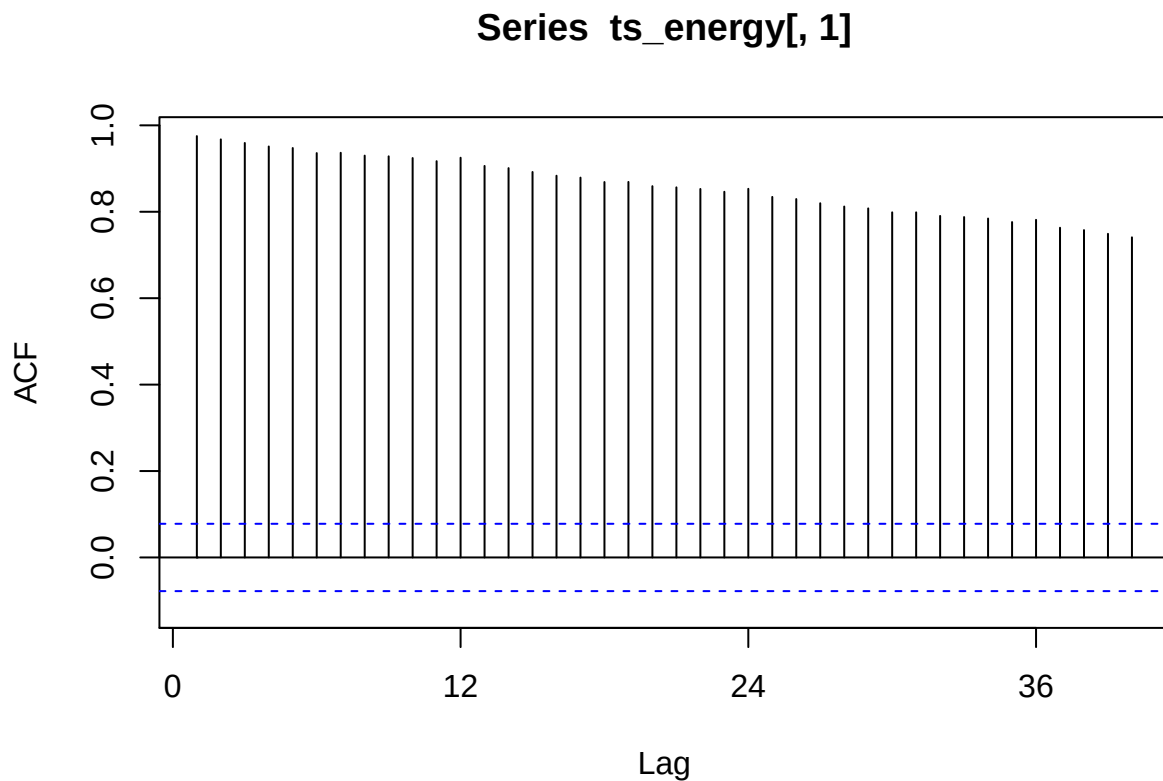
## Total Biomass Energy Production	-0.13473742
## Total Renewable Energy Production	-0.05842436
## Hydroelectric Power Consumption	1.00000000

The matrix shows the highest correlations with the same variable (as expected, 1 indicating perfectly correlated). The highest correlation is a positive correlation between biomass energy production and renewable energy production (0.96). The negatives indicate a negative correlation, but are all much closer to 0 ($<|0.2|$). This indicates that there is not a strong correlation for either of those correlated variables with hydroelectric power consumption. This makes sense visually based on the time series plots - hydroelectric plotted uniquely.

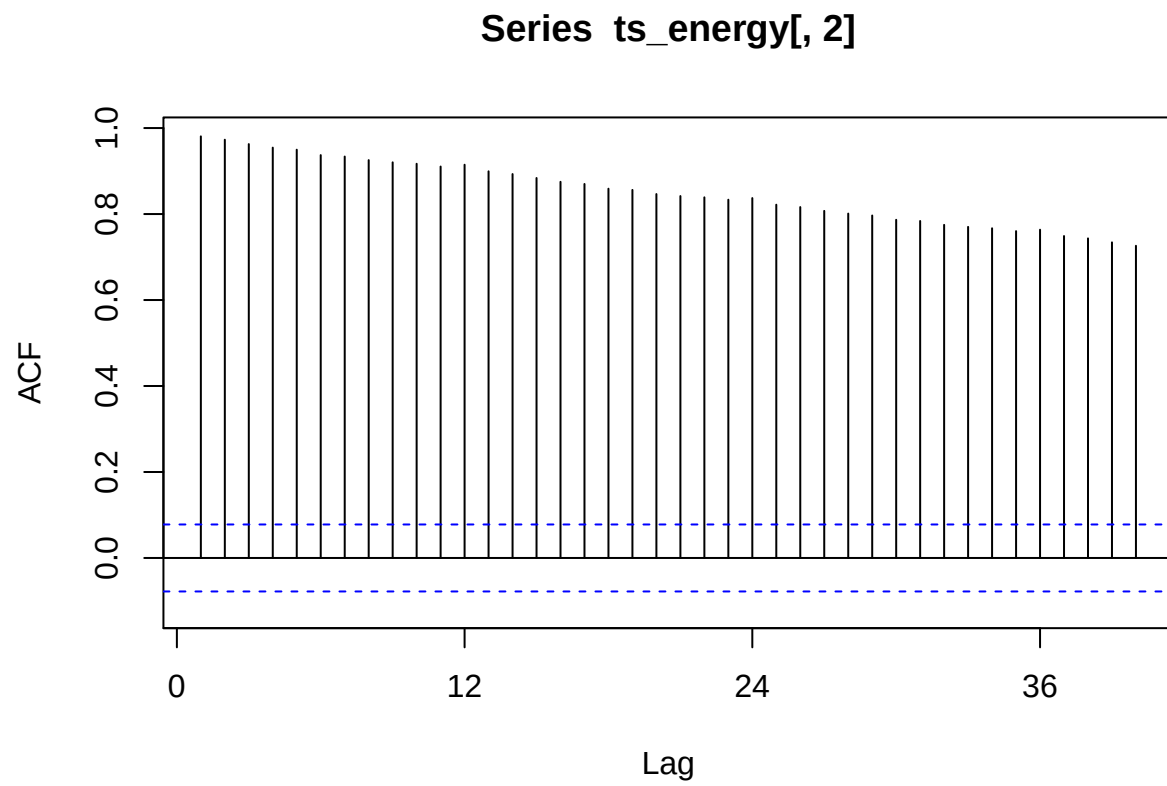
Question 6

Compute the autocorrelation function from lag 1 up to lag 40 for these three variables. What can you say about these plots? Do the three of them have the same behavior?

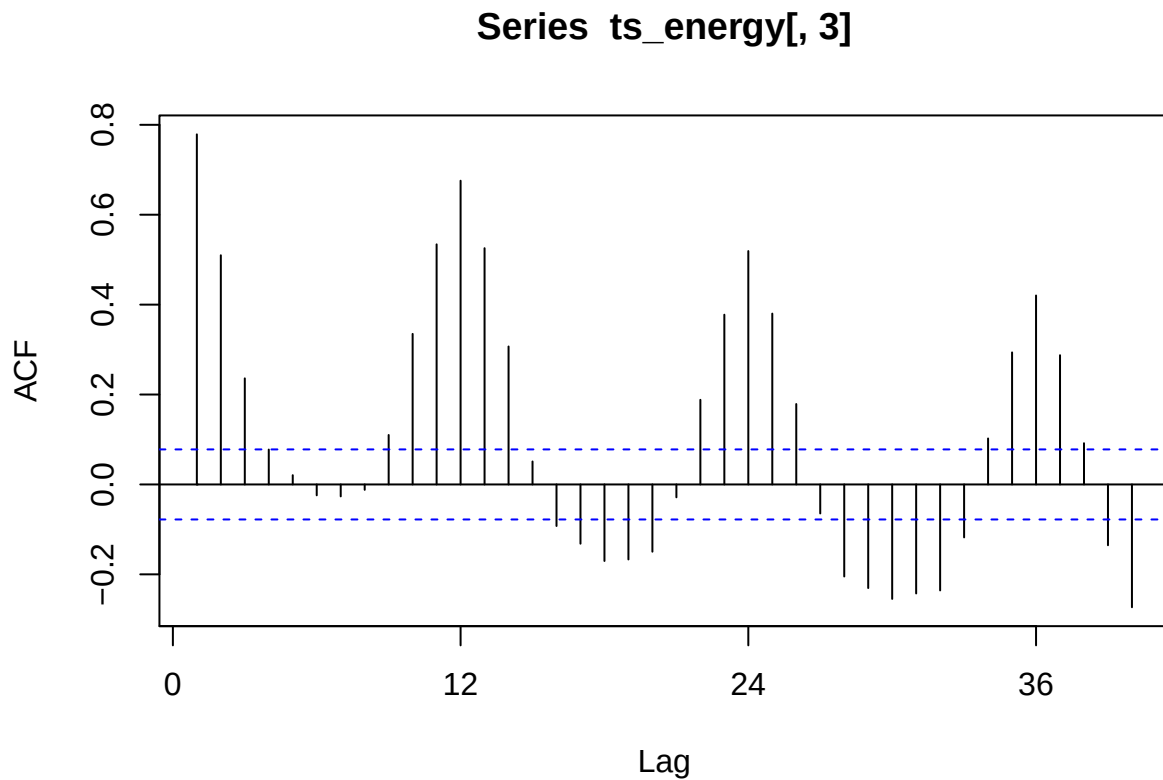
```
Biomass_ACF= Acf(ts_energy[,1],lag.max=40,plot=TRUE)
```



```
Renewables_ACF=Acf(ts_energy[,2], lag.max=40, plot=TRUE)
```

```
Hydroelectric_ACF = Acf(ts_energy[,3], lag.max=40, plot=TRUE)
```



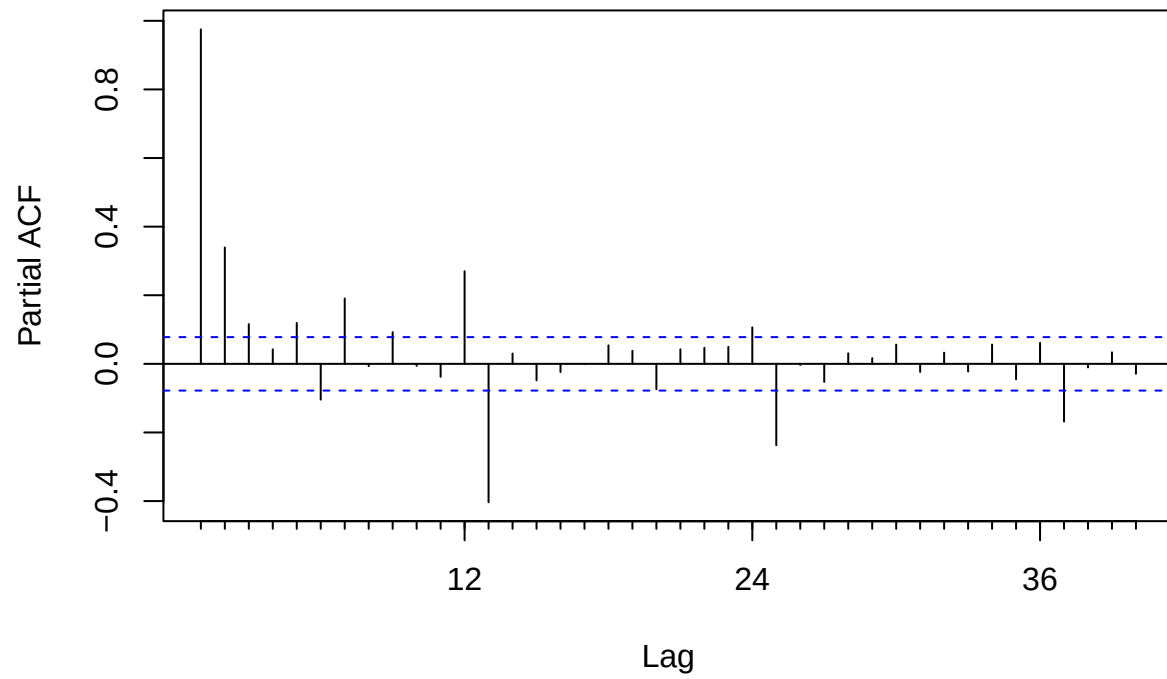
For biomass and renewables, Lags over time seem very similar very gradual drop-off toward insignificant. This likely indicates that past values of the time series' are correlated to future values. The hydroelectric ACF also gradually declines so past values likely also are correlated to future values but has a totally different shape. The hydroelectric has positive and negative lags, while the first two are all positive. Hydro likely is a seasonally dependant variable while the other two time series are not seasonally dependant.

Question 7

Compute the partial autocorrelation function from lag 1 to lag 40 for these three variables. How these plots differ from the ones in Q6?

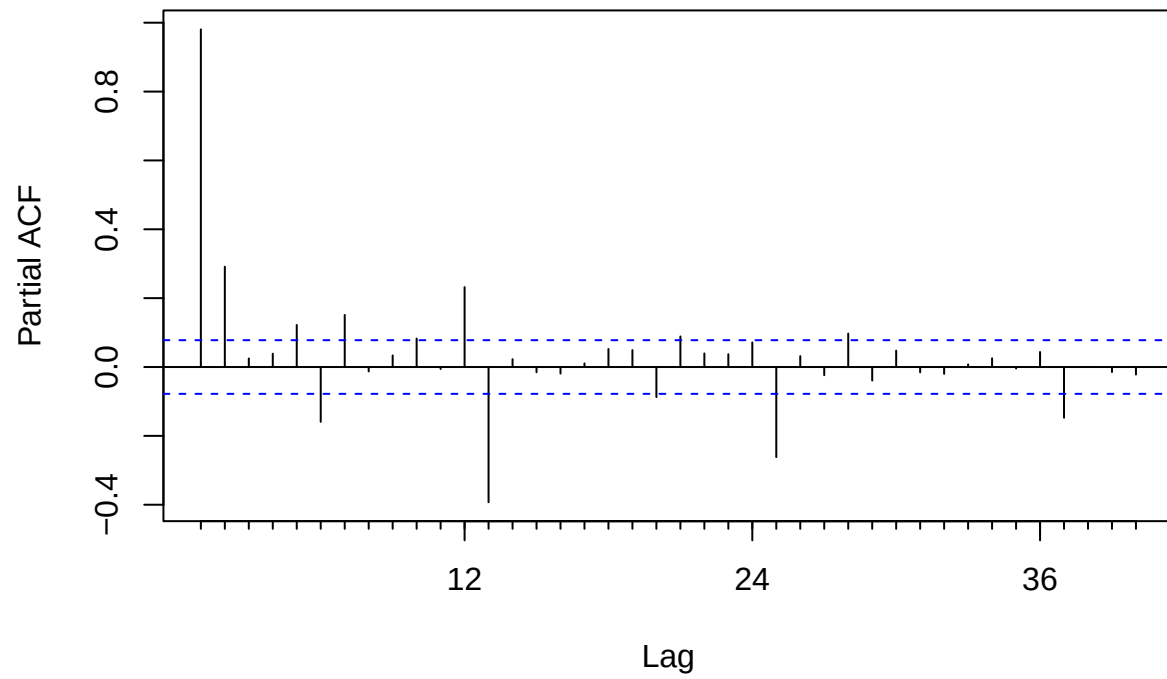
```
Biomass_PACF= Pacf(ts_energy[,1],lag.max=40,plot=TRUE)
```

Series ts_energy[, 1]

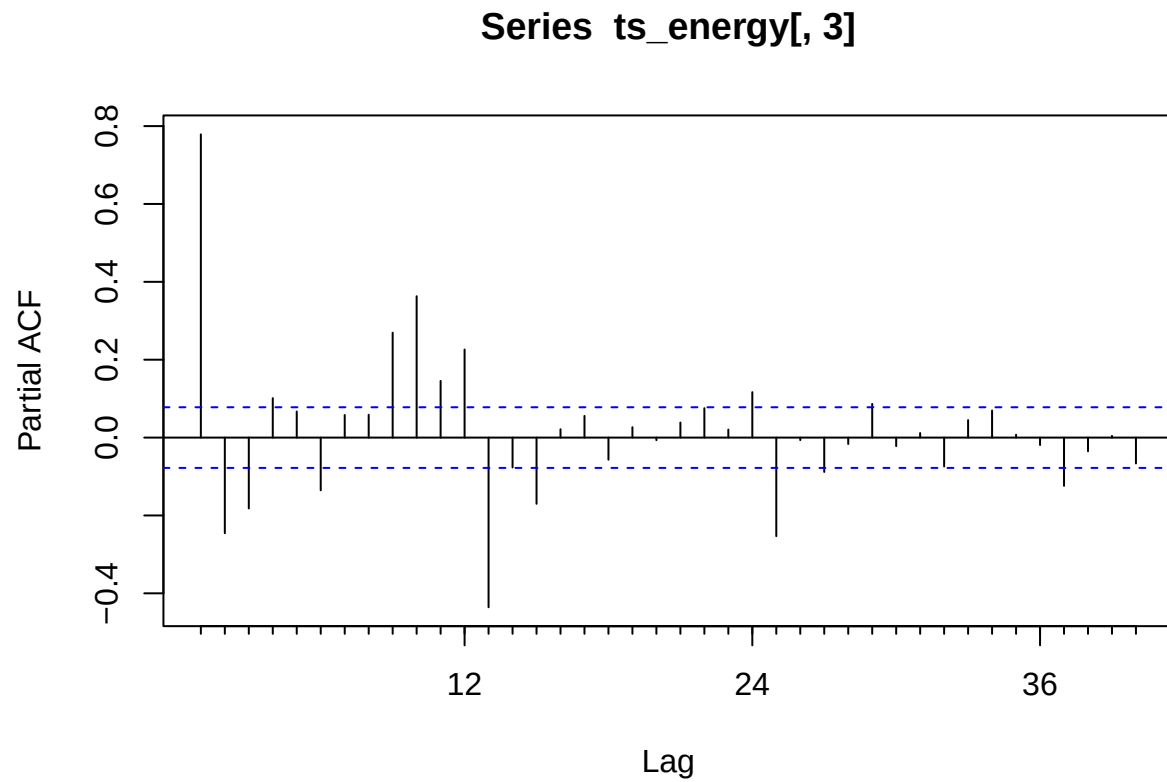


```
Renewables_PACF=Pacf(ts_energy[,2], lag.max=40, plot=TRUE)
```

Series ts_energy[, 2]



```
Hydroelectric_PACF = Pacf(ts_energy[,3], lag.max=40, plot=TRUE)
```



The PACF plots differ from the ACF plots in that PACF removes intermediate lag effects (direct relationship between Y and lag). Here we see much fewer significant lag terms. After the first few initial lags, both renewables and biomass energy have some significant negative lags (saw plotted earlier as periods where linear growth stalled out). Hydroelectric has more significant lags.